

第 4 章 線性規劃問題求解法：單行法

題目 4.4-5 (a) (b)

4.4-5.

(a) Set $x_1 = x_2 = x_3 = 0$.

$$(0) \quad Z - 2x_1 - 4x_2 - 3x_3 = 0$$

$$(1) \quad x_1 + 3x_2 + 2x_3 + x_4 = 80 \Rightarrow x_4 = 80$$

$$(2) \quad 3x_1 + 4x_2 + 2x_3 + x_5 = 60 \Rightarrow x_5 = 60$$

$$(3) \quad 2x_1 + x_2 + 2x_3 + x_6 = 40 \Rightarrow x_6 = 40$$

Optimality Test: The coefficients of all nonbasic variables are negative, so the solution $(0, 0, 0, 80, 60, 40)$ is not optimal.

Choose x_2 as the entering basic variable, since it has the largest coefficient.

$$(1) \quad x_1 + 3x_2 + 2x_3 + x_4 = 80 \Rightarrow x_4 = 80 - 3x_2 \Rightarrow x_2 \leq 26.67$$

$$(2) \quad 3x_1 + 4x_2 + 2x_3 + x_5 = 60 \Rightarrow x_5 = 60 - 4x_2 \Rightarrow x_2 \leq 15 \leftarrow \text{minimum}$$

$$(3) \quad 2x_1 + x_2 + 2x_3 + x_6 = 40 \Rightarrow x_6 = 40 - x_2 \Rightarrow x_2 \leq 40$$

We choose x_5 as the leaving basic variable. Set $x_1 = x_5 = x_3 = 0$.

$$(0) \quad Z + x_1 - x_3 + x_5 = 60$$

$$(1) \quad -1.25x_1 + 0.5x_3 + x_4 - 0.75x_5 = 35 \Rightarrow x_4 = 35$$

$$(2) \quad 0.75x_1 + x_2 + 0.5x_3 - 0.25x_5 = 15 \Rightarrow x_2 = 15$$

$$(3) \quad 1.25x_1 + 1.5x_3 - 0.25x_5 + x_6 = 25 \Rightarrow x_6 = 25$$

Optimality Test: The coefficient of x_3 is negative, so the solution $(0, 15, 0, 35, 0, 25)$ is not optimal.

Let x_3 be the entering basic variable.

$$(1) \quad -1.25x_1 + 0.5x_3 + x_4 - 0.75x_5 = 35 \Rightarrow x_4 = 35 - 0.5x_3 \Rightarrow x_3 \leq 70$$

$$(2) \quad 0.75x_1 + x_2 + 0.5x_3 + 0.25x_5 = 15 \Rightarrow x_2 = 15 - 0.5x_3 \Rightarrow x_3 \leq 30$$

$$(3) \quad 1.25x_1 + 1.5x_3 - 0.25x_5 + x_6 = 25 \Rightarrow x_6 = 25 - 1.5x_3 \Rightarrow x_3 \leq 16.67 \leftarrow \min$$

We choose x_6 as the leaving basic variable. Set $x_1 = x_5 = x_6 = 0$.

$$(0) \quad Z + 1.83x_1 + 0.83x_5 + 0.67x_6 = 76.67$$

$$(1) \quad -1.67x_1 + x_4 - 0.67x_5 - 0.33x_6 = 26.67 \Rightarrow x_4 = 26.67$$

$$(2) \quad 0.33x_1 + x_2 + 0.33x_5 - 0.33x_6 = 6.67 \Rightarrow x_2 = 6.67$$

$$(3) \quad 0.83x_1 + x_3 - 0.17x_5 + 0.67x_6 = 16.67 \Rightarrow x_3 = 16.67$$

Optimality Test: All of the coefficients are positive, so the solution $(0, 6.67, 16.67, 26.67, 0, 0)$ is optimal. $Z^* = 76.67$.

(b) Optimal solution: $(x_1^*, x_2^*, x_3^*) = (0, 6.67, 16.67)$ and $Z^* = 76.67$

Bas Var	Eq No	Z	Coefficient of						Right side
			X1	X2	X3	X4	X5	X6	
Z	0	1	-2	-4	-3	0	0	0	0
X4	1	0	1	3	2	1	0	0	80
X5	2	0	3	4*	2	0	1	0	60
X6	3	0	2	1	2	0	0	1	40

Bas Var	Eq No	Z	Coefficient of						Right side
			X1	X2	X3	X4	X5	X6	
Z	0	1	1	0	-1	0	1	0	60
X4	1	0	-1.25	0	0.5	1	-0.75	0	35
X2	2	0	0.75	1	0.5	0	0.25	0	15
X6	3	0	1.25	0	1.5*	0	-0.25	1	25

Bas Var	Eq No	Z	Coefficient of						Right side
			X1	X2	X3	X4	X5	X6	
Z	0	1	1.833	0	0	0	0.833	0.667	76.67
X4	1	0	-1.67	0	0	1	-0.67	-0.33	26.67
X2	2	0	0.333	1	0	0	0.333	-0.33	6.667
X3	3	0	0.833	0	1	0	-0.17	0.667	16.67

題目 4.4-6 (a) (b)

4.4-6.

(a) Optimal Solution: $(x_1^*, x_2^*, x_3^*) = (0, \frac{4}{3}, \frac{4}{3})$ and $Z^* = 14\frac{2}{3}$

0)	Z-	3	X ₁ -	5	X ₂ -	6	X ₃ +	0	X ₄ +	0	X ₅ +	0	X ₆ +	0	X ₇	=	0
1)		2	X ₁ +	1	X ₂ +	1	X ₃ +	1	X ₄ +	0	X ₅ +	0	X ₆ +	0	X ₇	=	4
2)		1	X ₁ +	2	X ₂ +	1	X ₃ +	0	X ₄ +	1	X ₅ +	0	X ₆ +	0	X ₇	=	4
3)		1	X ₁ +	1	X ₂ +	2	X ₃ +	0	X ₄ +	0	X ₅ +	1	X ₆ +	0	X ₇	=	4
4)		1	X ₁ +	1	X ₂ +	1	X ₃ +	0	X ₄ +	0	X ₅ +	0	X ₆ +	1	X ₇	=	3

$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0, x_6 \geq 0, x_7 \geq 0.$

0)	Z+	0	X ₁ -	2	X ₂ +	0	X ₃ +	0	X ₄ +	0	X ₅ +	3	X ₆ +	0	X ₇	=	12
1)		1.5	X ₁ +	0.5	X ₂ +	0	X ₃ +	1	X ₄ +	0	X ₅ -	0.5	X ₆ +	0	X ₇	=	2
2)		0.5	X ₁ +	1.5	X ₂ +	0	X ₃ +	0	X ₄ +	1	X ₅ -	0.5	X ₆ +	0	X ₇	=	2
3)		0.5	X ₁ +	0.5	X ₂ +	1	X ₃ +	0	X ₄ +	0	X ₅ +	0.5	X ₆ +	0	X ₇	=	2
4)		0.5	X ₁ +	0.5	X ₂ +	0	X ₃ +	0	X ₄ +	0	X ₅ -	0.5	X ₆ +	1	X ₇	=	1

$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0, x_6 \geq 0, x_7 \geq 0.$

0)	Z+	0.67	X ₁ +	0	X ₂ +	0	X ₃ +	0	X ₄ +	1.33	X ₅ +	2.33	X ₆ +	0	X ₇	=	14.6667
1)		1.333	X ₁ +	0	X ₂ +	0	X ₃ +	1	X ₄ +	-0.33	X ₅ +	-0.33	X ₆ +	0	X ₇	=	1.33333
2)		0.333	X ₁ +	1	X ₂ +	0	X ₃ +	0	X ₄ +	0.67	X ₅ +	-0.33	X ₆ +	0	X ₇	=	1.33333
3)		0.333	X ₁ +	0	X ₂ +	1	X ₃ +	0	X ₄ +	-0.33	X ₅ +	0.67	X ₆ +	0	X ₇	=	1.33333
4)		0.333	X ₁ +	0	X ₂ +	0	X ₃ +	0	X ₄ +	-0.33	X ₅ +	-0.33	X ₆ +	1	X ₇	=	0.33333

$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0, x_6 \geq 0, x_7 \geq 0.$

(b) Optimal Solution: $(x_1^*, x_2^*, x_3^*) = (0, \frac{4}{3}, \frac{4}{3})$ and $Z^* = 14\frac{2}{3}$

Bas Var	Eq No	Z	Coefficient of							Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	X ₇	
Z	0	1	-3	-5	-6	0	0	0	0	0
X ₄	1	0	2	1	1	1	0	0	0	4
X ₅	2	0	1	2	1	0	1	0	0	4
X ₆	3	0	1	1	2	0	0	1	0	4
X ₇	4	0	1	1	1	0	0	0	1	3

Bas Var	Eq No	Z	Coefficient of							Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	X ₇	
Z	0	1	0	-2	0	0	0	3	0	12
X ₄	1	0	1.5	0.5	0	1	0	-0.5	0	2
X ₅	2	0	0.5	1.5	0	0	1	-0.5	0	2
X ₃	3	0	0.5	0.5	1	0	0	0.5	0	2
X ₇	4	0	0.5	0.5	0	0	0	-0.5	1	1

Bas Var	Eq No	Z	Coefficient of							Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	X ₇	
Z	0	1	0.6667	0	0	0	1.3333	2.3333	0	14.6667
X ₄	1	0	1.3333	0	0	1	-0.333	-0.333	0	1.33333
X ₂	2	0	0.3333	1	0	0	0.6667	-0.333	0	1.33333
X ₃	3	0	0.3333	0	1	0	-0.333	0.6667	0	1.33333
X ₇	4	0	0.3333	0	0	0	-0.333	-0.333	1	0.33333

題目 4.6-2 (a) (b) (c) (d) (e) (f)

4.6-2.

(a) - (b) Initial artificial BF solution: $(0, 0, 0, 0, 300, 300)$

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	-10M	-4M	-5M	-7M			
X ₅	1	0	-4	-2	-3	-5	0	0	-600M
X ₅	1	0	2	3	4	2	1	0	300
X ₆	2	0	8	1	1	5	0	1	300

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	-2.75M	-3.75M	-0.75M			1.25M	-225M
X ₅	1	0	0	-1.5	-2.5	-2.5	0	+0.5	+150
X ₅	1	0	0	2.75	3.75	0.75	1	-0.25	225
X ₁	2	0	1	0.125	0.125	0.625	0	0.125	37.5

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	0	0.3333	0	-2	+0.667	+0.333	300
X ₃	1	0	0	0.7333	1	0.2	0.2667	-0.067	60
X ₁	2	0	1	0.0333	0	0.6	-0.033	0.1333	30

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	3.3333	0.4444	0	0	+0.556	+0.778	400
X ₃	1	0	-0.333	0.7222	1	0	0.2778	-0.111	50
X ₄	2	0	1.6667	0.0556	0	1	-0.056	0.2222	50

Optimal Solution: $(x_1^*, x_2^*, x_3^*, x_4^*) = (0, 0, 50, 50)$ and $Z^* = 400$

(c) - (d) - (e) - (f) Initial artificial BF solution: $(0, 0, 0, 0, 300, 300)$

Phase 1:

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	-10	-4	-5	-7	0	0	-600
X ₅	1	0	2	3	4	2	1	0	300
X ₆	2	0	8	1	1	5	0	1	300

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	0	-2.75	-3.75	-0.75	0	1.25	-225
X ₅	1	0	0	2.75	3.75	0.75	1	-0.25	225
X ₁	2	0	1	0.125	0.125	0.625	0	0.125	37.5

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	0	0	0	0	1	1	0
X ₃	1	0	0	0.7333	1	0.2	0.2667	-0.067	60
X ₁	2	0	1	0.0333	0	0.6	-0.033	0.1333	30

Phase 2:

Bas Var	Eq No	Z	Coefficient of				Right Side
			X ₁	X ₂	X ₃	X ₄	
Z	0	1	0	0.3333	0	-2	300
X ₃	1	0	0	0.7333	1	0.2	60
X ₁	2	0	1	0.0333	0	0.6	30

Bas Var	Eq No	Z	Coefficient of				Right Side
			X ₁	X ₂	X ₃	X ₄	
Z	0	1	3.3333	0.4444	0	0	400
X ₃	1	0	-0.333	0.7222	1	0	50
X ₄	2	0	1.6667	0.0556	0	1	50

Optimal Solution: $(x_1^*, x_2^*, x_3^*, x_4^*) = (0, 0, 50, 50)$ and $Z^* = 400$

題目 4.6-3 (a) (b) (c)

4.6-3.

(a) maximize $-Z = -2x_1 - 3x_2 - x_3$
 subject to $-x_1 - 4x_2 - 2x_3 \leq -8$
 $-3x_1 - 2x_2 \leq -6$
 $x_1, x_2, x_3 \geq 0$

(b) Optimal Solution: $(x_1^*, x_2^*, x_3^*) = (0.8, 1.8, 0)$ and $Z^* = 7$

Bas Var	Eq No	Z	Coefficient of							Right Side
			x_1	x_2	x_3	x_4	x_5	x_6	x_7	
Z	0	1	-4M	-6M	-2M					
x_6	1	0	+2	+3	+1	1M	1M	0	0	-14M
x_7	2	0	1	4	2	-1	0	1	0	8
			3	2	0	0	-1	0	1	6

Bas Var	Eq No	Z	Coefficient of							Right Side
			x_1	x_2	x_3	x_4	x_5	x_6	x_7	
Z	0	1	-2.5M		1M	-0.5M		1.5M		-2M
x_2	1	0	+1.25	0	-0.5	+0.75	1M	-0.75	0	-6
x_7	2	0	0.25	1	0.5	-0.25	0	0.25	0	2
			2.5	0	-1	0.5	-1	-0.5	1	2

Bas Var	Eq No	Z	Coefficient of							Right Side
			x_1	x_2	x_3	x_4	x_5	x_6	x_7	
Z	0	1	0	0	0	0.5	0.5	1M	1M	-7
x_2	1	0	0	1	0.6	-0.3	0.1	-0.5	-0.5	1.8
x_1	2	0	1	0	-0.4	0.2	-0.4	0.3	-0.1	0.8
								-0.2	0.4	0.8

Pivoting x_3 for x_2 gives an alternate optimal BF solution, $(2, 0, 3)$.

(c) Optimal Solution: $(x_1^*, x_2^*, x_3^*) = (0.8, 1.8, 0)$ and $Z^* = 7$

Phase 1:

Bas Var	Eq No	Z	Coefficient of							Right Side
			x_1	x_2	x_3	x_4	x_5	x_6	x_7	
Z	0	1	-4	-6	-2	1	1	0	0	-14
x_6	1	0	1	4	2	-1	0	1	0	8
x_7	2	0	3	2	0	0	-1	0	1	6

Bas Var	Eq No	Z	Coefficient of							Right Side
			x_1	x_2	x_3	x_4	x_5	x_6	x_7	
Z	0	1	-2.5	0	1	-0.5	1	1.5	0	-2
x_2	1	0	0.25	1	0.5	-0.25	0	0.25	0	2
x_7	2	0	2.5	0	-1	0.5	-1	-0.5	1	2

Bas Var	Eq No	Coefficient of								Right Side
		Z	X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	X ₇	
Z	0	1	0	0	0	0	0	1	1	0
X ₂	1	0	0	1	0.6	-0.3	0.1	0.3	-0.1	1.8
X ₁	2	0	1	0	-0.4	0.2	-0.4	-0.2	0.4	0.8

Phase 2:

Bas Var	Eq No	Coefficient of					Right Side
		Z	X ₁	X ₂	X ₃	X ₄	
Z	0	1	0	0	^{~0} -5e-20	0.5	0.5
X ₂	1	0	0	1	0.6	-0.3	0.1
X ₁	2	0	1	0	-0.4	0.2	-0.4

Pivoting x_3 for x_2 gives an alternate optimal BF solution, (2, 0, 3).

題目 4.6-7 (a) (b) (c) (d) (e) (f)

4.6-7.

(a) Initial artificial BF solution: $(0, 0, 0, 0, 20, 50)$

Bas Var	Eq No	Z	Coefficient of						Right Side
			X_1	X_2	X_3	X_4	X_5	X_6	
Z	0	1	-3M	-2M	-2M				
X_5	1	0	-2	-5	-3	1M	0	0	-70M
X_6	2	0	1	-2	1	-1	1	0	20
			2	4	1	0	0	1	50

(b) Optimal Solution: $(x_1^*, x_2^*, x_3^*) = (0, 0, 50)$ and $Z^* = 150$

Bas Var	Eq No	Z	Coefficient of						Right Side
			X_1	X_2	X_3	X_4	X_5	X_6	
Z	0	1		-8M	1M	-2M	3M		-10M
X_1	1	0	0	-9	-1	-2	+2	0	+40
X_6	2	0	1	-2	1	-1	1	0	20
			0	8	-1	2	-2	1	10

Bas Var	Eq No	Z	Coefficient of						Right Side
			X_1	X_2	X_3	X_4	X_5	X_6	
Z	0	1					1M	1M	
X_1	1	0	0	0	-2.125	0.25	-0.25	+1.125	51.25
X_2	2	0	1	0	0.75	-0.5	0.5	0.25	22.5
			0	1	-0.125	0.25	-0.25	0.125	1.25

Bas Var	Eq No	Z	Coefficient of						Right Side
			X_1	X_2	X_3	X_4	X_5	X_6	
Z	0	1					1M	1M	
X_3	1	0	2.8333	0	0	-1.167	+1.167	+1.833	115
X_2	2	0	1.3333	0	1	-0.667	0.6667	0.3333	30
			0.1667	1	0	0.1667	-0.167	0.1667	5

Bas Var	Eq No	Z	Coefficient of						Right Side
			X_1	X_2	X_3	X_4	X_5	X_6	
Z	0	1						1M	
X_3	1	0	4	7	0	0	1M	+3	150
X_4	2	0	2	4	1	0	0	1	50
			1	6	0	1	-1	1	30

(c) Initial artificial BF solution: $(0, 0, 0, 0, 20, 50)$

Phase 1:

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	-3	-2	-2	1	0	0	-70
X ₅	1	0	1	-2	1	-1	1	0	20
X ₆	2	0	2	4	1	0	0	1	50

(d)

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	0	-8	1	-2	3	0	-10
X ₁	1	0	1	-2	1	-1	1	0	20
X ₆	2	0	0	8	-1	2	-2	1	10

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	0	0	0	0	1	1	0
X ₁	1	0	1	0	0.75	-0.5	0.5	0.25	22.5
X ₂	2	0	0	1	-0.125	0.25	-0.25	0.125	1.25

(e) - (f) Optimal Solution: $(x_1^*, x_2^*, x_3^*) = (0, 0, 50)$ and $Z^* = 150$

Phase 2:

Bas Var	Eq No	Z	Coefficient of				Right Side
			X ₁	X ₂	X ₃	X ₄	
Z	0	1	0	0	-2.125	0.25	51.25
X ₁	1	0	1	0	0.75	-0.5	22.5
X ₂	2	0	0	1	-0.125	0.25	1.25

Bas Var	Eq No	Z	Coefficient of				Right Side
			X ₁	X ₂	X ₃	X ₄	
Z	0	1	2.8333	0	0	-1.167	115
X ₃	1	0	1.3333	0	1	-0.667	30
X ₂	2	0	0.1667	1	0	0.1667	5

Bas Var	Eq No	Z	Coefficient of				Right Side
			X ₁	X ₂	X ₃	X ₄	
Z	0	1	4	7	0	0	150
X ₃	1	0	2	4	1	0	50
X ₄	2	0	1	6	0	1	30

題目 4.6-9 (a) (b)

4.6-9.

(a) Optimal Solution: $(x_1^*, x_2^*, x_3^*) = (0, 15, 15)$ and $Z^* = 90$

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	-5M	-4M	-8M				
X ₅	1	0	+3	+2	+4	1M	0	0	-180M
X ₆	2	0	2	1	3	0	1	0	60
			3	3	5	-1	0	1	120

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	0.333M	-1.333M			2.667M		-20M
X ₃	1	0	+0.333	+0.667	0	1M	-1.333	0	-80
X ₆	2	0	0.6667	0.3333	1	0	0.3333	0	20
			-0.333	1.3333	0	-1	-1.667	1	20

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	0.5	0	0	0.5	-0.5	-0.5	-90
X ₃	1	0	0.75	0	1	0.25	0.75	-0.25	15
X ₂	2	0	-0.25	1	0	-0.75	-1.25	0.75	15

(b) Optimal Solution: $(x_1^*, x_2^*, x_3^*) = (0, 15, 15)$ and $Z^* = 90$

Phase 1:

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	-5	-4	-8	1	0	0	-180
X ₅	1	0	2	1	3	0	1	0	60
X ₆	2	0	3	3	5	-1	0	1	120

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	0.3333	-1.333	0	1	2.6667	0	-20
X ₃	1	0	0.6667	0.3333	1	0	0.3333	0	20
X ₆	2	0	-0.333	1.3333	0	-1	-1.667	1	20

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	-3e-20	0	0	0	1	1	0
X ₃	1	0	0.75	0	1	0.25	0.75	-0.25	15
X ₂	2	0	-0.25	1	0	-0.75	-1.25	0.75	15

Phase 2:

Bas Var	Eq No	Z	Coefficient of				Right Side
			X ₁	X ₂	X ₃	X ₄	
Z	0	1	0.5	0	0	0.5	-90
X ₃	1	0	0.75	0	1	0.25	15
X ₂	2	0	-0.25	1	0	-0.75	15

$$\text{Min } Z = 3X_1 + 2X_2 + 4X_3 \Rightarrow \text{Min } Z = 3X_1 + 2X_2 + 4X_3 + M\bar{X}_5 + M\bar{X}_6$$

$$2X_1 + X_2 + 3X_3 = 60 \Rightarrow 2X_1 + X_2 + 3X_3 + \bar{X}_5 = 60$$

$$3X_1 + 3X_2 + 5X_3 \geq 120 \Rightarrow 3X_1 + 3X_2 + 5X_3 - X_4 + \bar{X}_6 = 120$$

$$\Rightarrow \text{Max } -Z = -3X_1 - 2X_2 - 4X_3 - M\bar{X}_5 - M\bar{X}_6$$

$$\begin{cases} 2X_1 + X_2 + 3X_3 + \bar{X}_5 = 60 \\ 3X_1 + 3X_2 + 5X_3 - X_4 + \bar{X}_6 = 120 \\ \bar{X}_5 = 60 - 2X_1 - X_2 - 3X_3 \\ \bar{X}_6 = 120 - 3X_1 - 3X_2 - 5X_3 + X_4 \end{cases}$$

$$\Rightarrow -Z = -3X_1 - 2X_2 - 4X_3 - M(60 - 2X_1 - X_2 - 3X_3) - M(120 - 3X_1 - 3X_2 - 5X_3 + X_4)$$

$$\Rightarrow -Z = -3X_1 - 2X_2 - 4X_3 - 60M + 2MX_1 + MX_2 + 3MX_3 - 120M + 3MX_1 + 3MX_2 + 5MX_3 - MX_4$$

$$\Rightarrow -Z = (5M-3)X_1 + (4M-2)X_2 + (8M-4)X_3 + MX_4 - 180M$$

4.6-9

P. 1

	Z	X_1	X_2	X_3	X_4	$\bar{X}_5$	$\bar{X}_6$	RHS
Z	-1	$-5M$ +3	$-4M$ +2	$-8M$ +4	M	0	0	$-180M$
$\bar{X}_5$	0	2	1	3	0	1	0	60
$\bar{X}_6$	0	3	3	5	-1	0	1	120
Z	-1	$\frac{1}{3}M$ $+\frac{1}{3}$	$-\frac{4}{3}M$ $+\frac{2}{3}$	0	M	$\frac{8}{3}M - \frac{4}{3}$	0	$-20M - 80$
X_3	0	$\frac{2}{3}$	$\frac{1}{3}$	1	0	$\frac{1}{3}$	0	20
$\bar{X}_6$	0	$-\frac{1}{3}$	$\frac{4}{3}$	0	-1	$-\frac{5}{3}$	1	20
Z	-1	0.5	0	0	0.5	$M-0.5$	$M-0.5$	-90
X_3	0	$\frac{3}{4}$	0	1	$\frac{1}{4}$	$\frac{3}{4}$	$-\frac{1}{4}$	15
X_2	0	$-\frac{1}{4}$	1	0	$-\frac{3}{4}$	$-\frac{5}{4}$	$\frac{3}{4}$	15

$$\Rightarrow (X_1, X_2, X_3, X_4, \bar{X}_5, \bar{X}_6)$$

$$= (0, 15, 15, 0, 0, 0)$$

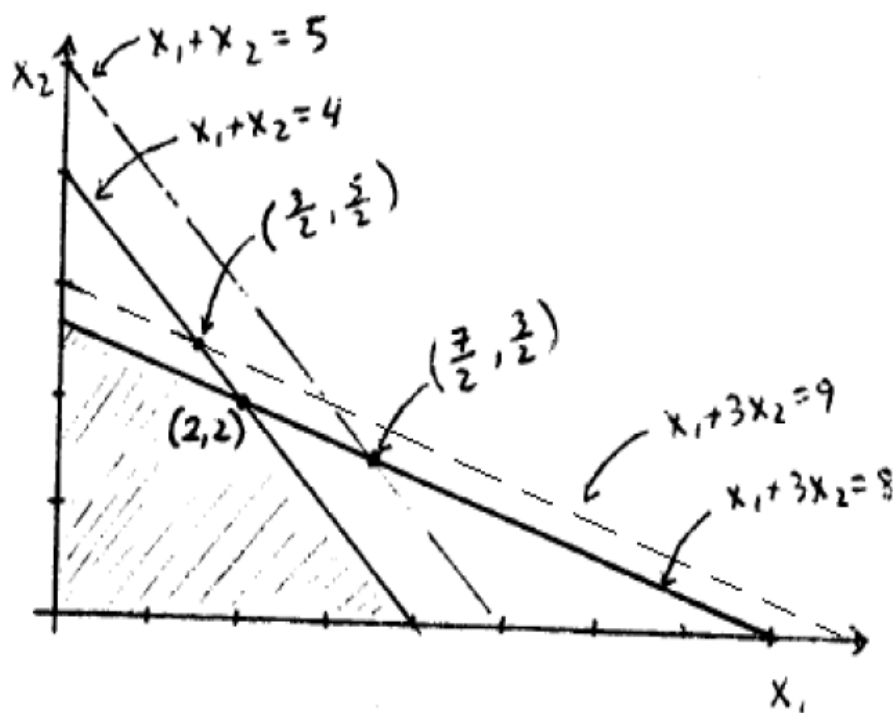
$$-Z = -90 \Rightarrow Z = 90$$

P.2

題目 4.7-2 (a) (b) (c) (d)

4.7-2.

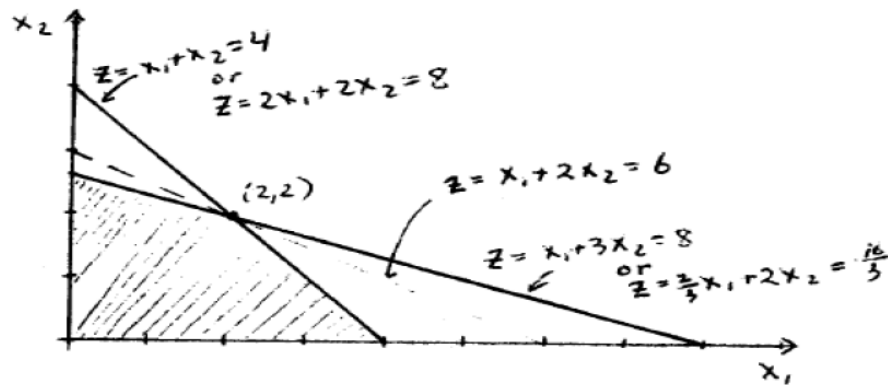
(a)



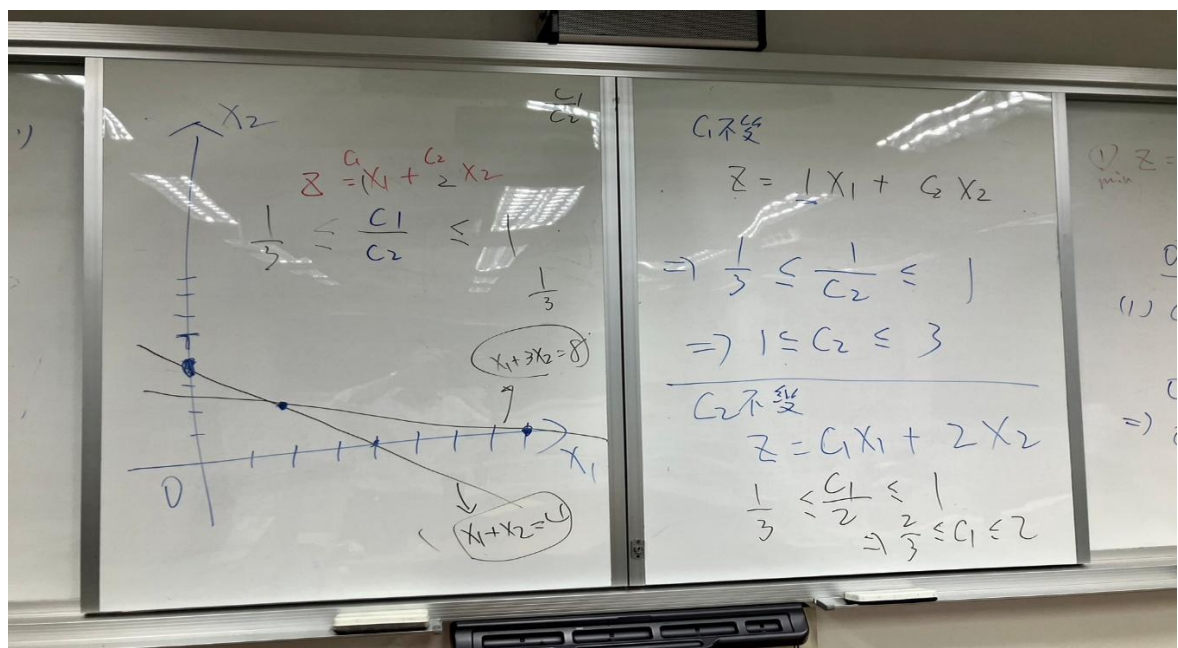
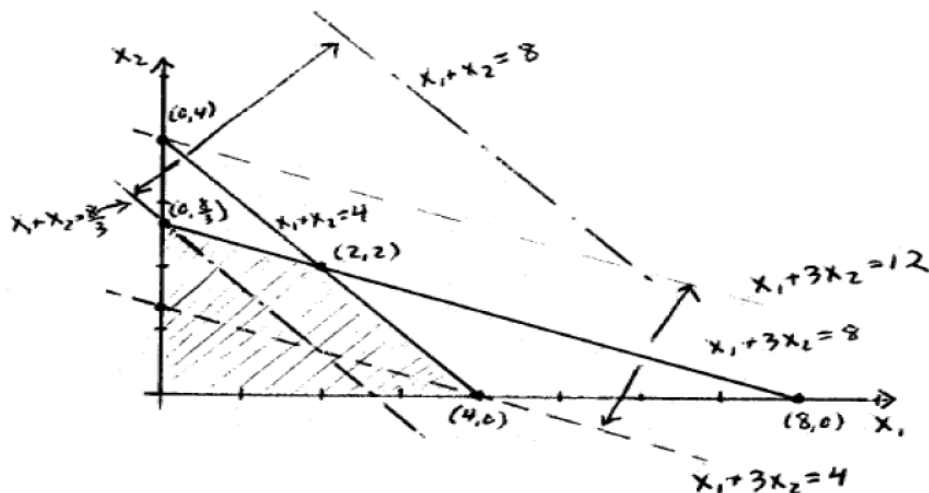
Constraint (1): $x_1 + 3x_2 \leq 8$: $x_1 + 3x_2 = 8 \Rightarrow x_1 = x_2 = 2$ and $Z = 6$
 $x_1 + 3x_2 = 9 \Rightarrow x_1 = 3/2, x_2 = 5/2$ and $Z = 13/2$
 $\Delta Z = 13/2 - 6 = 1/2 = y_1^*$
 Constraint (2): $x_1 + x_2 \leq 4$: $x_1 + x_2 = 4 \Rightarrow x_1 = x_2 = 2$ and $Z = 6$
 $x_1 + x_2 = 5 \Rightarrow x_1 = 7/2, x_2 = 3/2$ and $Z = 13/2$
 $\Delta Z = 13/2 - 6 = 1/2 = y_2^*$

(b) From (a), we see that the right-hand sides $b_1 = 8$ and $b_2 = 4$ are sensitive parameters. The graph in part (a) shows that both constraints are active (binding) at the optimal solution, so all the coefficients $a_{11} = 1$, $a_{12} = 3$, $a_{21} = 1$, and $a_{22} = 1$ are sensitive parameters, too. As will be seen in (c), the objective coefficients $c_1 = 1$ and $c_2 = 2$ are not sensitive parameters.

(c) Observe that the optimal solution remains the same for $2/3 \leq c_1 \leq 2$ (with $c_2 = 2$ fixed) and $1 \leq c_2 \leq 3$ (with $c_1 = 1$ fixed)



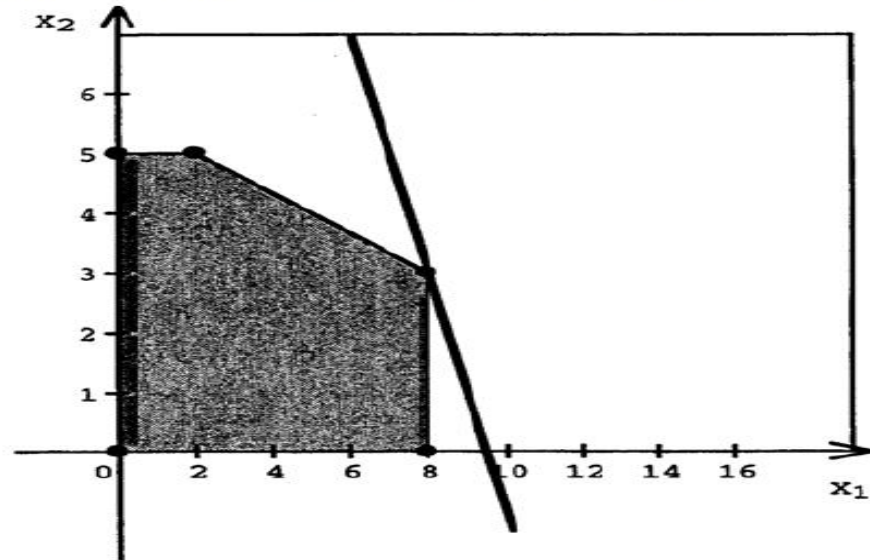
(d) The dashed lines "- - -" in the graph below suggest that the CP solution ranges from $(4, 0)$ to $(0, 4)$ when $4 \leq b_1 \leq 12$. Outside this range, the CP solution becomes infeasible. The dashed lines "- · -" represent the second constraint for different right-hand side values. They suggest that the CP solution ranges from $(0, 8/3)$ to $(0, 8)$ when $8/3 \leq b_2 \leq 8$. Hence, the allowable ranges are $4 \leq b_1 \leq 12$ and $8/3 \leq b_2 \leq 8$.



題目 4.7-3

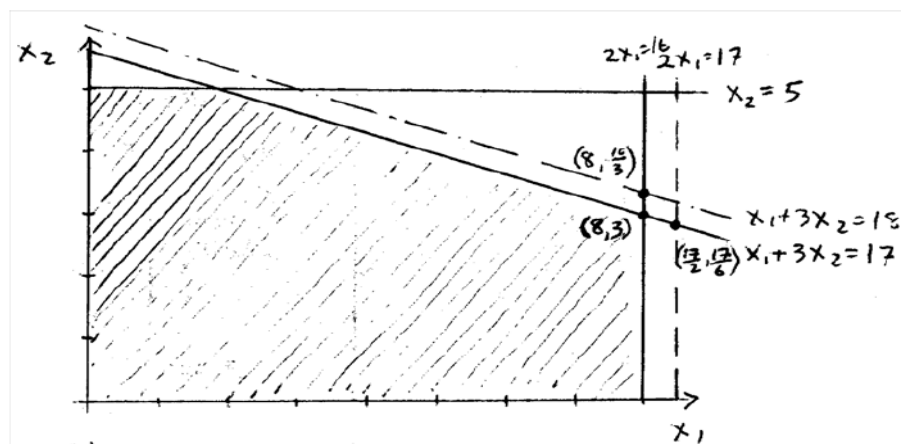
4.7-3.

(a) Optimal Solution: $(x_1^*, x_2^*) = (8, 3)$ and $Z^* = 38$



Corner Point	Z
$(8, 3)$	38^*
$(8, 0)$	32
$(2, 5)$	18
$(0, 5)$	10
$(0, 0)$	0

(b)



Increasing resource 1 to 17 units increases Z to $4(8.5) + 2(2.83) = 39.67$, so $\Delta Z = y_1^* = 1.67$.

Increasing resource 2 to 18 units increases Z to $4(8) + 2(3.33) = 38.33$, so $\Delta Z = y_2^* = 0.67$.

The third constraint is not binding, so $y_3^* = 0$.

(c) To increase Z by 15, resource 1 should be increased by $\frac{15}{y_1^*} = \frac{15}{1.67} \approx 9$. Solving the LP problem with resource 1 set to $16 + 9 = 25$ returns the result $Z = 53$.

題目 4.7-4 (a) (b)

4.7-4.

(a) Optimal Solution: $(x_1^*, x_2^*, x_3^*) = (0.5, 0, 4.5)$ and $Z^* = 14$

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	-1	7	-3	0	0	0	0
X ₄	1	0	2	1	-1	1	0	0	4
X ₅	2	0	4	-3	0	0	1	0	2
X ₆	3	0	-3	2	1	0	0	1	3

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	-10	13	0	0	0	3	9
X ₄	1	0	-1	3	0	1	0	1	7
X ₅	2	0	4	-3	0	0	1	0	2
X ₃	3	0	-3	2	1	0	0	1	3

Bas Var	Eq No	Z	Coefficient of						Right Side
			X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	
Z	0	1	0	5.5	0	0	2.5	3	14
X ₄	1	0	0	2.25	0	1	0.25	1	7.5
X ₁	2	0	1	-0.75	0	0	0.25	0	0.5
X ₃	3	0	0	-0.25	1	0	0.75	1	4.5

(b) The shadow prices for the three resources are given by the reduced costs (in the objective function) for the corresponding slack variables. These values are circled in the table above. The shadow prices for resources 1, 2 and 3 are 0, 2.5 and 3 respectively. They represent the rate at which the objective function value z increases as the corresponding resource is increased. For instance, increasing resource 3 by one unit increases Z by 3, provided that no other constraints cause any trouble.